GAP-10T Minimal One-Connection GR Audit: An Architecture-Level No-Go and the Exact Completion Fork

Ing. David Jaroš

26 July 2026

AI provenance — Tier B. AI-assisted; machine-verified against named sources or verifiers; individual attestation is not claimed. Details: `AI_PROVENANCE.md`.

Abstract

We combine two previously proved UBT results: the concurrent-vector obstruction for a torsion-free tetrad generated by the pure Lorentz pair, and the local right inverse for an arbitrary tetrad obtained by adding composite contortion. The combination yields an architecture-level dichotomy. If the one connection inside $D\Theta$ is also the physical metric connection and the physical branch is exact torsion-free general relativity, then the generated tetrad admits a concurrent vector and generic GR geometries, including the non-flat Schwarzschild exterior, are excluded. If nonzero contortion is retained to represent arbitrary tetrads, the same physical connection has nonzero torsion and the theory is not exact torsion-free GR. Therefore the minimal architecture with one and the same physical Lorentz connection cannot simultaneously provide universal local single-Θ representability and exact generic GR. The result does not exclude UBT gravity; it proves that an exact-GR completion must relax at least one assumption, most economically by separating a nonpropagating jet connection from the physical Levi–Civita connection, or else by accepting a genuinely torsionful gravity theory.

1 Assumptions

Let

$$E_\mu = \mathcal{N}_0^{-1/2} D^\Omega_\mu \Theta, \qquad D^\Omega_\mu X = \partial_\mu X + \Omega_\mu X + X\Omega^\ddagger_\mu, \tag{1}$$

with E_μ a nondegenerate Lorentz tetrad. The minimal *one-connection* interpretation imposes:

1. Ω is the only Lorentz connection in the defining jet;
2. the same Ω is the spin lift of the physical metric-compatible affine connection that enters curvature and matter transport;
3. no independent relative left/right connection acts on Θ.

Exact ordinary GR further requires the physical connection to be the Levi–Civita connection of E on the spinless vacuum branch.

2 The dichotomy

Theorem 1 (Minimal one-connection exact-GR no-go)**.** *Under the assumptions above, the same one-connection architecture cannot have both of the following properties:*

1. *every smooth Lorentzian tetrad is locally representable by a single Θ through Eq.* (1)*;*

2. *the physical vacuum branch is exact torsion-free GR for generic metrics.*

Proof. If the physical branch is torsion-free, metric compatibility fixes $\Omega = \mathring{\Omega}(E)$. The previously proved concurrent-vector theorem then applies to the Lorentz-real projection of Θ: there is a spacetime vector V such that

$$\mathring{\nabla}_\mu V^\nu = \delta_\mu{}^\nu, \qquad \mathcal{L}_V g_{\mu\nu} = 2g_{\mu\nu}, \qquad R^\rho{}_{\sigma\mu\nu} V^\sigma = 0. \tag{2}$$

Generic torsion-free Einstein geometries do not admit this structure; in particular, the non-flat Schwarzschild vacuum exterior with $M \neq 0$ admits no proper homothety of Eq. (2). Hence exact generic GR fails.

Conversely, the local representer theorem constructs a right inverse for every smooth tetrad by using

$$\Omega = \mathring{\Omega}(E) + K(E, \Theta), \qquad K \neq 0 \text{ in general.} \tag{3}$$

For a metric-compatible connection, contortion and torsion are in one-to-one algebraic correspondence. Therefore $K \neq 0$ makes the same physical affine connection torsionful. It may define an Einstein–Cartan or other modified branch, but it is not the Levi–Civita connection of exact ordinary GR. Thus the two desired properties cannot hold simultaneously under the same one-connection interpretation. □

Corollary 2 (What must be relaxed)**.** *An exact generic-GR completion of the single-*Θ *tetrad map must relax at least one of the following:*

1. *identify the jet connection with the physical spacetime connection;*
2. *restrict the derivative to the pure Lorentz pair of Eq.* (1)*;*
3. *require exact torsion-free GR rather than a torsionful modification;*
4. *require local representability of an arbitrary tetrad by one* Θ*.*

3 Controlled completion fork

The smallest exact-GR architecture suggested by the theorem is

$$\widehat{\Omega} = \mathring{\Omega}(E) + K_J[E, \Theta], \qquad E_\mu = \mathcal{N}_0^{-1/2} D_\mu^{\widehat{\Omega}} \Theta, \tag{4}$$

for the defining jet, while physical curvature and ordinary matter transport use

$$\Omega_{\text{phys}} = \mathring{\Omega}(E), \qquad T_{\text{phys}} = 0. \tag{5}$$

A companion theorem gives an explicit local composite Lorentz plus relative- central jet right inverse for every non-null Lorentz-real projection of Θ. It must not be interpreted as physical spacetime torsion. The remaining issue is dynamical selection of that right inverse and of the tetrad, not local kinematic existence.

The alternative is to keep one connection and interpret the result as a physical torsionful theory. That route requires observational bounds and an action that dynamically selects the composite contortion.

4 What remains for a complete GR derivation

Even after choosing the split exact-GR architecture, two independent lemmas remain:

1. **Jet dynamics lemma:** derive from the canonical action the selection of the explicit split-jet right inverse and of $E[\Theta]$, and prove that the relative central/Lorentz jet components are nonpropagating.
2. **Curvature-action lemma:** derive the Hilbert–Palatini/Einstein– Hilbert term, including its sign and Newton coefficient, from the canonical UBT action. The locked quadratic kinetic scalar is only a volume term and cannot supply R by algebraic rewriting.

Once these are proved, standard Palatini variation and the already proved connection-reconstruction theorem give the Einstein–Λ equations on the torsion-free physical branch. Until then, the endpoint is a controlled conditional completion rather than an unconditional derivation from the original minimal action.

Exact status

GAP-10T-MINIMAL-ONE-CONNECTION-GR: CLOSED AS NO-GO [L1]. The same one metric-compatible physical Lorentz connection cannot provide both universal local single-Θ representability and exact generic torsion-free GR.

GAP-10T-JET-KIN: CLOSED LOCALLY [L1]. The split exact-GR architecture has an explicit local composite jet right inverse while physical curvature remains Levi–Civita.

GAP-10T-JET-AUX: CLOSED [L1]. The follow-up multiplier completion makes the split-jet variables algebraic, nonpropagating, and on-shell decoupled on every non-null patch.

GAP-10T-JET-CONSTRAINT-SELECTION: CLOSED AS NO-GO [L1]. Surjectivity of the right inverse prevents the pure constraint from selecting a physical tetrad.

GAP-10T-JET-DYN: NARROWED. The canonical action has not yet selected the physical tetrad and jet representative, or derived a separate non-surjective metric effective law; null-patch/global continuation and the constrained mode measure remain.

GAP-10D: NARROWED, NOT CLOSED. The physical Einstein endpoint is conditionally standard once its curvature term is present; the canonical origin and coefficient of that term remain to be derived.

References

[1] R. M. Wald, *General Relativity*, University of Chicago Press (1984).

[2] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, "General Relativity with Spin and Torsion: Foundations and Prospects," *Rev. Mod. Phys.* **48**, 393–416 (1976).